Journal of Machine Learning Research (2026) 1-13 Submitted 11/26; Published

Neurons or Subspaces? Calibration-Sensitive Refusal in Open Instruction Models

Varadaraj Acharya VARADARAJ277@GMAIL.COM
Independent Researcher
Udupi, Karnataka, India

Editor: Action Editor

Abstract

Safety-aligned language models refuse harmful requests, but the mechanism is contested: is refusal implemented by a sparse set of MLP neurons, or by a linear direction in the residual stream? We study this question on four open instruction models using minimally paired harmful/benign prompts from JailbreakBench, a brutal HarmBench transfer set, and a content-matched style control. Ablating a mean-difference direction removes refusal in Qwen3.5-0.8B (53.3% → 0.0%) and DeepSeek-R1-Distill-1.5B (26.7% → 6.7%), and transfers to HarmBench (42.0% → 0.0% and 31.1% → 6.7%). LFM3-1.7B is calibration-sensitive: the matched direction reduces refusal only from 86.7% to 73.3%, while a direction estimated on a larger natural corpus reaches 82.0% → 21.0% at a KL cost of 1.05. Qwen3.5-4B is the strongest refusal model and the hardest case: 96.7% → 50.0% at KL 1.16, with HarmBench 93.3% → 42.0%. Removing the direction at a single layer is much weaker than removing it at every layer, so the effect is distributed rather than localised. Neuron clamping is model-dependent: 80 neurons across four layers reduce refusal to 20.0% in Qwen3.5-0.8B, leave R1-Distill unchanged at 26.7%, reduce LFM3 to 36.7%, and leave Qwen3.5-4B at 96.7%. Adding the direction induces refusal but at a large distribution cost (KL up to 4.45). We conclude that refusal is a distributed, calibration-sensitive linear mechanism—not a single neuron, and not a universal layer or direction.

Keywords: refusal, mechanistic interpretability, activation steering, calibration, safety alignment

1 Introduction

Instruction-tuned language models are trained to refuse harmful requests while remaining helpful on benign ones. The safety behaviour is remarkably consistent across model families, yet its internal implementation is still debated. One view holds that refusal is a *sparse circuit*: a small set of MLP neurons that fire on harmful intent and route generation into a refusal template. The other holds that refusal is a *linear direction* in the residual stream, spread across layers, which can be projected out or added at inference time.

The linear view has strong recent support. Arditi et al. (2024) identify a single difference-in-means direction in thirteen chat models and show that erasing it prevents refusal while adding it induces refusal on harmless instructions. This is a striking result, but three questions remain open.

First, *is the direction an artefact of prompt style?* The standard evaluation contrasts AdvBench harmful requests with Alpaca instructions. The two sets differ in topic, length,

©2026 Varadaraj Acharya.

License: CC-BY 4.0, see https://creativecommons.org/licenses/by/4.0/. Attribution requirements are provided at http://jmlr.org/papers/v/.html.

and mood: harmful prompts are usually imperatives ("Write a ransomware script"), while benign prompts are often questions ("How do I change a tire?"). A mean-difference direction can therefore encode request style as much as harmfulness.

Second, *is the direction universal?* A direction estimated on one corpus may not transfer to another, and the literature rarely reports how sensitive the estimate is to the calibration distribution.

Third, *do neurons matter?* Single-neuron ablation studies often test fewer than 1% of the MLP at a single, often final, layer. That is a narrow test of a broad hypothesis.

We address these questions with a controlled study on four heterogeneous open models: a hybrid linear-attention VLM (Qwen3.5-0.8B), a reasoning-distilled Qwen2 model (DeepSeek-R1-Distill-1.5B), a hybrid short-conv/attention model (LFM3-1.7B), and a higher-refusal hybrid VLM (Qwen3.5-4B). Our evaluation uses: (i) *minimally paired* JailbreakBench prompts, where each harmful request has a benign counterpart with the same template, topic and length; (ii) a *brutal external* HarmBench set, never seen during calibration; (iii) a *style control* that renders the same benign prompts as imperatives and as questions; and (iv) a *neuron budget* sweep from 1 to 5% of the MLP, including multi-layer clamping.

Contributions.

1. A matched-pair evaluation of refusal directions that removes the imperative-vs-question and topic confounds of AdvBench-vs-Alpaca designs, with a content-matched style direction as a negative control.
2. Evidence that the refusal direction is *calibration-sensitive*: it is strong in Qwen3.5-0.8B and R1-Distill, but in LFM3-1.7B the matched direction is weak while a natural-corpus direction is strong, at a higher KL cost.
3. A brutal transfer test: a direction estimated on moderate JBB pairs ablates refusal on HarmBench for three of four models.
4. A neuron-budget study showing that single-layer top-k clamping is weak everywhere, while 80 neurons across four layers can substantially reduce refusal in some models and not others.
5. A fully reproducible pipeline, including disjoint splits, random-direction and random-neuron controls, Wilson intervals, and KL cost.

2 Related Work

Refusal directions and activation steering. Arditi et al. (2024) compute a difference-in-means direction at the last instruction token and show that directional ablation bypasses refusal across thirteen models. Activation addition (Turner et al., 2023) and contrastive activation addition (Panickssery et al., 2023) use comparable interventions for style and sentiment. Our ablation follows the same all-layer scope; our contribution is the matched-pair evaluation, the calibration analysis, and the neuron comparison.

Geometry of refusal. Marshall et al. (2024) show that refusal is better described by an affine function than by a pure direction, and that directional ablation alone can produce incoherent outputs. Wollschläger et al. (2025) identify multiple independent refusal directions

and multi-dimensional concept cones. Our results are consistent with a dominant direction but not with exclusivity: the estimate is calibration-sensitive and the intervention has a measurable distribution cost.

Neuron-level safety. Chen et al. (2024) identify a small fraction of safety neurons and show that patching them restores much of the safety behaviour. This is compatible with our finding that single neurons are insufficient but multi-layer neuron clamping can matter; the two literatures differ in budget, layer choice, and causal score.

Reliability of steering. Tan et al. (2024) show that steering vectors can be unreliable across prompts and can have opposite-of-intended effects. We therefore report held-out prompt-level rates, random controls, and KL rather than a single prompt.

Evaluation benchmarks. JailbreakBench (Chao et al., 2024) provides paired harmful and benign behaviours with matching categories. HarmBench (Mazeika et al., 2024) provides a large, more severe behaviour set. AdvBench (Zou et al., 2023) is the common harmful corpus; Alpaca is the common benign one. Our design uses JBB for the matched test and HarmBench for external severity, and reserves AdvBench/Alpaca for layer selection.

3 Method

3.1 Refusal direction

For each layer l, let μ_l^+ and μ_l^- be the mean residual-stream activations at the last prompt token for harmful and benign calibration prompts. The refusal direction is

$$r_l = \frac{\mu_l^+ - \mu_l^-}{\|\mu_l^+ - \mu_l^-\|_2}. \tag{1}$$

The decision token is the final token of the chat-formatted prompt. For Qwen3.5 and LFM3 we disable thinking so that the first generated token is the answer; for R1-Distill we prefill an empty `</think>` block for the same reason.

3.2 Layer selection

The layer is chosen on an independent natural validation split (96 harmful / 48 harmless AdvBench/Alpaca prompts), not on the matched test distribution. For each candidate layer we compute (i) the mean refusal score on harmful validation prompts after all-layer ablation, (ii) the mean refusal score on harmless prompts after adding the direction, and (iii) the KL divergence on harmless prompts. We prune the last 20% of layers and choose

$$L^\star = \arg \min_{l\,:\,\mathrm{KL}_l \le 0.1} s_l^{\mathrm{abl}}. \tag{2}$$

This follows the selection logic of Arditi et al. (2024) and avoids the "largest separation norm" heuristic, which is biased toward the final layer.

3.3 Interventions

Ablation. We project the unit direction out of the embedding output and every block output:

$$h \leftarrow h - (h^\top r)\, r. \tag{3}$$

The *single-layer* control applies the same projection only at $L^\star$.

Addition. We add the direction at $L^\star$ to all token positions,

$$h \leftarrow h + \alpha r, \qquad \alpha = m\, \|\mu^+_{L^\star} - \mu^-_{L^\star}\|_2, \tag{4}$$

with $m \in \{0, 0.5, 1, 1.5, 2\}$. This matches the scale used by Arditi et al. (2024).

3.4 Neuron attribution and clamping

For layer $L^\star$ and intermediate neuron j, we score

$$S_j = \big(\mathbb{E}^+[a_j] - \mathbb{E}^-[a_j]\big)\, \big(r^\top_{L^\star} W_{\text{down}}[:, j]\big), \tag{5}$$

where a_j is the pre-down-projection MLP activation. We clamp the top-k neurons to zero at every token position. We test $k \in \{1, 5, 20, 100, 500, 5\%\}$ of the MLP and compare against three random neuron sets. The multi-layer variant splits a budget across $\{L^\star{-}2, \ldots, L^\star{+}1\}$, scoring each layer with the same direction and clamping the top neurons per layer.

3.5 Metrics

The refusal score at the last prompt token is

$$s = \log p(\mathcal{R}) - \log\big(1 - p(\mathcal{R})\big), \tag{6}$$

where $\mathcal{R}$ is the set of refusal-initiating first tokens (`I`, `As`, `Sorry`, `No`) and $p(\mathcal{R})$ is their total probability mass. A prompt counts as a refusal when $s > 0$ (equivalently, $p(\mathcal{R}) > 0.5$). We report Wilson 95% intervals. KL is the mean divergence of the next-token distribution on harmless prompts versus the unmodified model.

3.6 Controls

Random direction. A random unit vector of the same norm is ablated and added, averaged over three draws.

Random neurons. Three random neuron sets of the same cardinality are clamped and averaged.

Style direction. The same benign JBB prompts are rendered as imperatives and as questions (content matched). The difference-in-means direction is a pure mood axis; we report its cosine with the refusal direction and the effect of ablating it.

External brutal set. A direction estimated on JBB is evaluated unchanged on HarmBench standard behaviours; the benign side is JBB benign prompts from the mapped domain.

Table 1: Models. All four models are evaluated; Qwen3.5-4B is the high-refusal case.

Model	Family	Layers	d	d_{MLP}	Status
Qwen3.5-0.8B	qwen3_5	24	1024	3584	complete
R1-Distill-1.5B	qwen2_r1	28	1536	8960	complete
LFM3-1.7B	lfm3	24	2048	7680	complete
Qwen3.5-4B	qwen3_5	32	2560	9216	complete

4 Experimental Setup

Models. Table 1 lists the models. Qwen3.5 is a hybrid linear/full-attention VLM; we use its text stack. LFM3 is a hybrid short-conv and attention model loaded with its remote code. R1-Distill is a standard Qwen2 transformer. Qwen3.5-4B is a larger hybrid VLM with a much higher baseline refusal rate (96.7% on JBB), which makes it the hardest case in the study.

Data. The direction is estimated on 50 matched JBB pairs; the layer is chosen on 96/48 AdvBench/Alpaca prompts; the main test is 30 held-out JBB pairs. The external set is 119 HarmBench standard behaviours, paired with JBB benign prompts from the same domain. The style control uses 50 benign JBB prompts in imperative and question form. Calibration, validation and test prompts are disjoint.

Compute. Apple M5 Pro (24 GB), MPS, float16, PyTorch 2.15.0 and Transformers 5.22.1. Prompts are batched with left padding; generation is greedy.

5 Results

5.1 Directional ablation

Table 2 and Figure 1 report the held-out matched-pair results. All-layer ablation removes refusal in Qwen3.5-0.8B (53.3% $\rightarrow$ 0.0%) and R1-Distill (26.7% $\rightarrow$ 6.7%). LFM3-1.7B is the exception: the matched direction reduces refusal only from 86.7% to 73.3%. Qwen3.5-4B is the strongest refusal model and the hardest case: 96.7% $\rightarrow$ 50.0% at KL 1.16, so the direction is real but the intervention is incomplete. Single-layer ablation is much weaker in every model (10.0%, 13.3%, 70.0%, 90.0%), so the mechanism is distributed across the residual stream rather than localised at $L^\star$. A random unit direction leaves the refusal score essentially unchanged ($\Delta \in [-0.09, +0.13]$), while the estimated direction lowers it substantially (for example, from $+0.13$ to -4.05 in Qwen3.5-0.8B).

5.2 Layer selection and calibration sensitivity

Figure 2 shows the validation ablation score by layer. The selected layers are 12 (Qwen3.5-0.8B), 15 (R1-Distill), 8 (LFM3-1.7B) and 9 (Qwen3.5-4B); the curves are smooth and the selected layer is not the last block. The choice is model-specific and does not collapse to a single depth.

Table 3 re-estimates the direction on the 96-prompt natural corpus and tests it on all 100 JBB pairs. Qwen3.5-0.8B and R1-Distill are stable (51.0% $\rightarrow$ 0.0% and 28.0% $\rightarrow$ 7.0%).

Table 2: Held-out JBB matched-pair results. Rates are percent with 95% Wilson intervals. Δ random is the change in mean refusal log-odds under a random unit direction (three draws). KL is on JBB benign prompts.

Model	Layer	Baseline (%)	Ablated (%)	Single-layer (%)	Δ random	KL
Qwen3.5-0.8B	12	53.3 [36.1, 69.8]	0.0 [0.0, 11.4]	10.0 [3.5, 25.6]	+0.10	0.568
R1-Distill-1.5B	15	26.7 [14.2, 44.4]	6.7 [1.8, 21.3]	13.3 [5.3, 29.7]	−0.02	0.367
LFM3-1.7B	8	86.7 [70.3, 94.7]	73.3 [55.6, 85.8]	70.0 [52.1, 83.3]	+0.13	0.193
Qwen3.5-4B	9	96.7 [83.3, 99.4]	50.0 [33.2, 66.8]	90.0 [74.4, 96.5]	−0.09	1.159

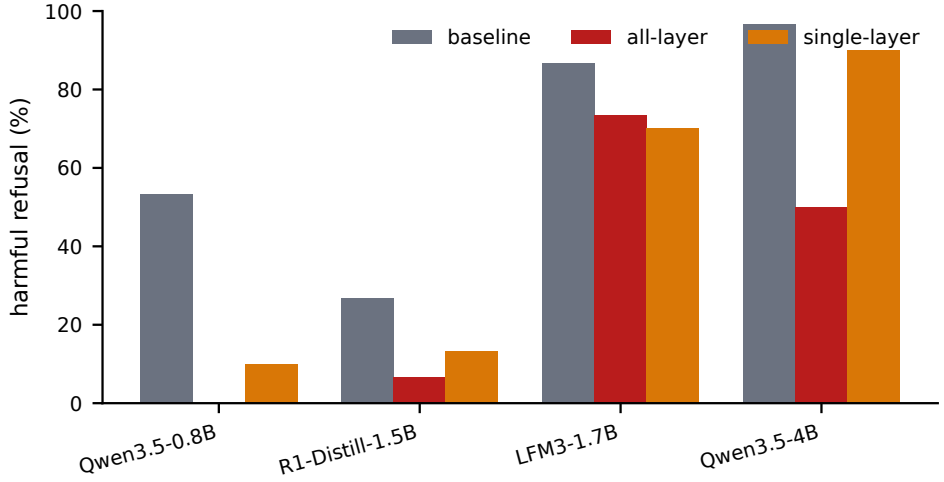


Figure 1: Harmful refusal before and after direction ablation, and under the single-layer control. The all-layer intervention is the causal effect; the single-layer bar shows that removing the direction at one layer is not enough.

Qwen3.5-4B is also stable but incomplete (93.0% → 40.0%, KL 1.70). LFM3-1.7B changes from a weak matched-direction effect (86.7% → 73.3%) to a strong natural-direction effect (82.0% → 21.0%), at KL 1.05 versus 0.19. The refusal direction in LFM3 is therefore real but its *estimate* depends on the calibration distribution.

5.3 Brutal transfer

Table 4 evaluates the JBB direction on 119 HarmBench standard behaviours. Refusal drops from 42.0% to 0.0% in Qwen3.5-0.8B and from 31.1% to 6.7% in R1-Distill, with small KL on the benign side (0.28). Qwen3.5-4B drops from 93.3% to 42.0%. LFM3-1.7B does not transfer (65.5% → 63.0%), consistent with its calibration sensitivity. The matched direction therefore generalises to much more severe prompts in three of four models, but not universally.

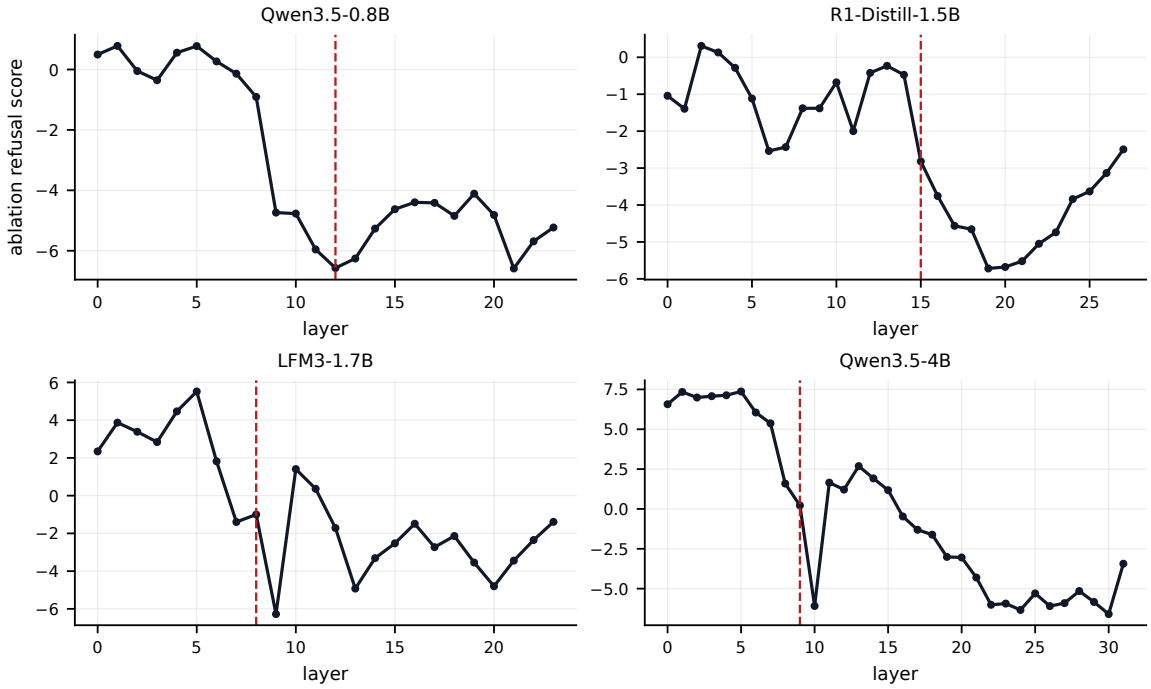


Figure 2: Validation ablation refusal score by layer. The dashed line marks the selected layer. Lower is better.

Table 3: Calibration sensitivity: direction re-estimated on 96 natural prompts and tested on all 100 JBB pairs. HarmBench is the brutal external set.

Model	Layer	Baseline (%)	Ablated (%)	KL	HarmBench (%)
Qwen3.5-0.8B	12	51.0	0.0	0.978	42.0 → 0.0
R1-Distill-1.5B	15	28.0	7.0	0.204	31.1 → 13.5
LFM3-1.7B	8	82.0	21.0	1.048	65.5 → 23.5
Qwen3.5-4B	9	93.0	40.0	1.700	93.3 → 34.4

5.4 Style control

Table 5 reports the imperative-vs-question direction. It is nearly orthogonal to the refusal direction in Qwen3.5-0.8B (cos = −0.16), R1-Distill (cos = −0.03) and Qwen3.5-4B (cos = −0.21), and ablating it does not reproduce the effect: Qwen3.5-0.8B falls only to 23.3%, while R1-Distill and LFM3-1.7B move to 93.3% and Qwen3.5-4B to 100.0% (refusal *increases*). The main direction is therefore not the mood axis.

Table 4: HarmBench transfer. The direction is computed on JBB pairs; the benign side is JBB benign prompts from the mapped domain.

Model	Brutal baseline (%)	Brutal ablated (%)	Benign baseline (%)	Benign ablated (%)	KL
Qwen3.5-0.8B	42.0	0.0	1.7	0.0	0.284
R1-Distill-1.5B	31.1	6.7	10.9	1.7	0.280
LFM3-1.7B	65.5	63.0	11.8	7.6	0.160
Qwen3.5-4B	93.3	42.0	19.3	2.5	0.753

Table 5: Style control: the same benign prompts as imperatives and as questions. cos is with the refusal direction at $L^\star$.

Model	cos	Baseline (%)	Style-ablated (%)	KL
Qwen3.5-0.8B	−0.156	53.3	23.3	0.388
R1-Distill-1.5B	−0.030	26.7	93.3	2.615
LFM3-1.7B	−0.093	86.7	93.3	0.355
Qwen3.5-4B	−0.210	96.7	100.0	0.731

5.5 Neuron ablation

Table 6 and Figure 3 show that single-layer neuron clamping is weak: at 5% of the MLP, refusal is 40.0% in Qwen3.5-0.8B (from a 53.3% baseline), 26.7% in R1-Distill (unchanged), 93.3% in LFM3-1.7B (from 86.7%), and 96.7% in Qwen3.5-4B (unchanged). The multi-layer budget is more effective: 80 neurons across four layers reduce refusal to 20.0% in Qwen3.5-0.8B (KL 0.13) and 36.7% in LFM3-1.7B (KL 0.77), but leave R1-Distill at 26.7% and Qwen3.5-4B at 96.7%; the 4B model needs 1840 neurons to reach 80.0% (KL 0.93). Random neuron sets do not match the top-k effect in Qwen3.5-0.8B or LFM3-1.7B. Neurons are neither irrelevant nor a universal switch; their causal role depends on the model and on the layer budget.

5.6 Addition

Table 7 and Figure 4 report the dose-response on harmless JBB prompts. At 1× the direction raises refusal to 26.7% in Qwen3.5-0.8B (KL 0.84), 23.3% in R1-Distill (KL 0.23), 73.3% in LFM3-1.7B (KL 1.90), and 73.3% in Qwen3.5-4B (KL 1.43). At 2× the rates are 66.7%, 53.3%, 63.3% and 100.0%, with KL 2.91, 0.93, 3.49 and 4.45. The direction is a real bidirectional knob, but it is not free: the KL cost is substantial, and LFM3-1.7B is non-monotonic. A high refusal rate is not by itself evidence of a clean intervention.

Table 6: Neuron ablation at about 5% of the MLP, and 80 neurons across four layers. Random score is the mean refusal log-odds under three random sets.

Model	k	% MLP	Top-k rate (%)	Top-k score	Random score	Multi-80 rate (%)	Multi-80 KL
Qwen3.5-0.8B	179	4.99	40.0	-0.36	+0.05	20.0	0.13
R1-Distill-1.5B	448	5.00	26.7	-1.01	-0.78	26.7	0.08
LFM3-1.7B	384	5.00	93.3	+2.06	+1.93	36.7	0.77
Qwen3.5-4B	460	4.99	96.7	+5.02	+5.64	96.7	0.46

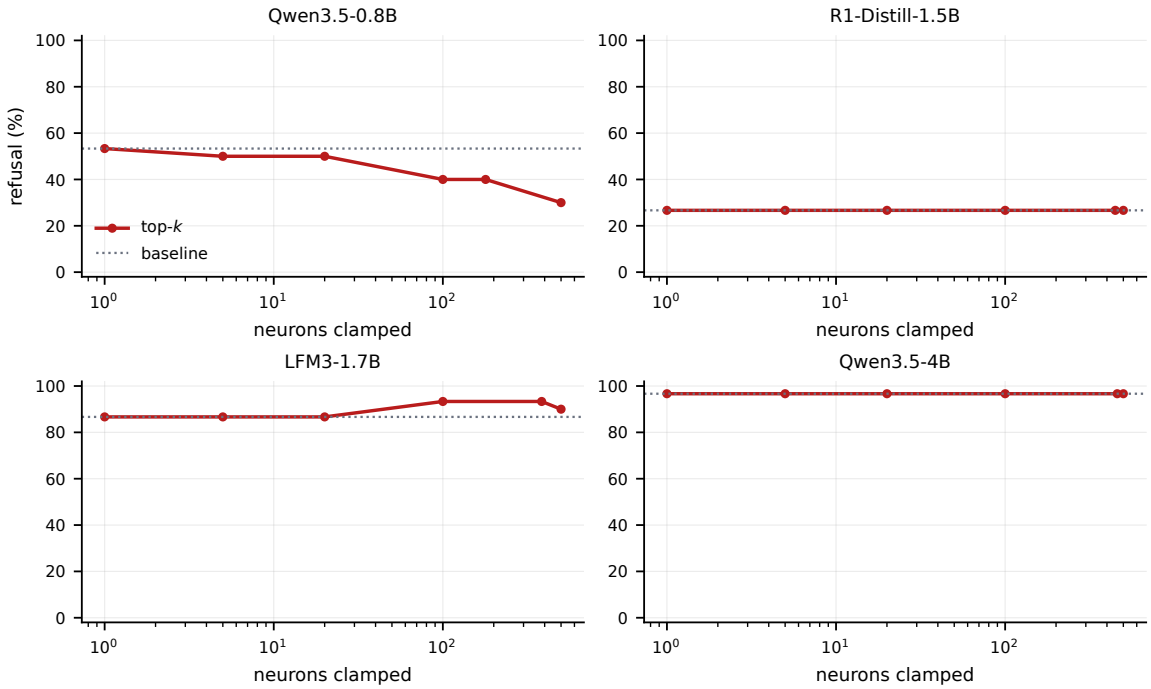


Figure 3: Refusal rate versus the number of clamped neurons at $L^\star$. The dotted line is the unmodified baseline. Single-layer clamping is weak; the multi-layer budget in Table 6 is the stronger intervention.

6 Discussion

A direction, but not a universal one. The mean-difference direction is a real causal object: it removes refusal when projected out, induces it when added, and a random direction does neither. But the estimate is calibration-sensitive. LFM3-1.7B is the clearest case: the matched direction barely moves refusal, while a natural-corpus direction moves it strongly at higher KL. Qwen3.5-4B is a different failure mode: the direction is strong but incomplete (96.7% $\rightarrow$ 50.0%) and costly (KL 1.16). The field should report calibration sensitivity and residual refusal rather than a single direction per model.

Table 7: Adding the direction to harmless JBB prompts. KL is versus the unmodified model.

Model	1.0× refusal (%)	1.0× KL	2.0× refusal (%)	2.0× KL
Qwen3.5-0.8B	26.7	0.84	66.7	2.91
R1-Distill-1.5B	23.3	0.23	53.3	0.93
LFM3-1.7B	73.3	1.90	63.3	3.49
Qwen3.5-4B	73.3	1.43	100.0	4.45

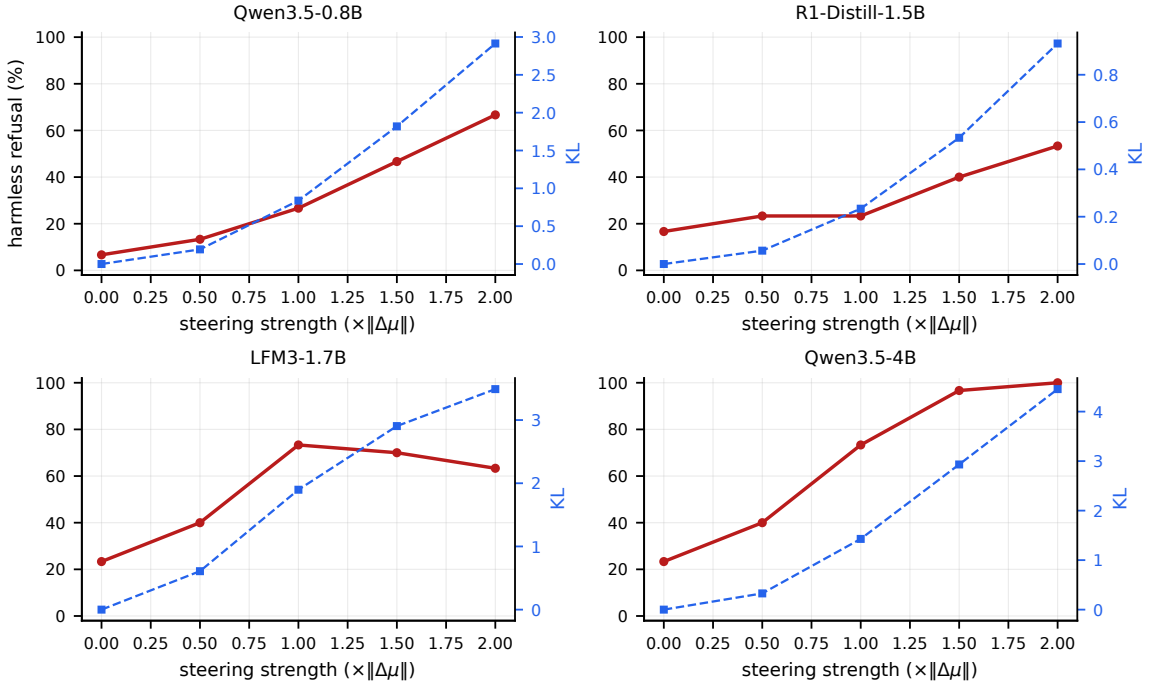


Figure 4: Refusal on harmless prompts (left axis) and KL versus the unmodified model (right axis) as the addition strength increases.

Neurons versus subspaces is not a binary. Single neurons are not the mechanism. At 1–20 neurons nothing happens anywhere. At 5% of the MLP, single-layer clamping is still weak; across four layers it becomes effective in Qwen3.5-0.8B and LFM3-1.7B, but not in R1-Distill or Qwen3.5-4B. The right question is not "neurons or subspaces?" but "which distributed subspace, at which budget, in which architecture?".

The cost of steering. Adding the direction induces refusal, but KL rises to 1–4.5 nats and the response can degenerate. A safety intervention that damages harmless behaviour

is not deployable, and a jailbreak that produces incoherent text is not a clean jailbreak. Reporting KL alongside refusal is necessary.

Implications for safety. If refusal is a thin linear offset, white-box safety is brittle: a few dozen calibration prompts and a forward hook can disable it. The brittleness is not uniform—architecture and calibration matter—but it is real, and it argues for defences that do not live only in a removable direction.

7 Limitations

- Four models and three architecture families. Qwen3.5-4B is the high-refusal case and its matched direction is incomplete (50% residual) with KL 1.16; a larger model may behave differently, and scaling the study is future work.
- Refusal is a proxy for harmfulness. We report log-odds and a string classifier, not an LLM judge or StrongREJECT score. A compliant answer can still be useless or incoherent; a refusal can be a topic restatement.
- Greedy decoding only; we do not estimate sampling variance.
- The matched test is 30 pairs. The external HarmBench set is category-matched, not minimally edited, so it tests severity generalisation rather than the confound.
- R1-Distill is evaluated with an empty thinking block; its refusal behaviour under normal chain-of-thought may differ.
- LFM3-1.7B is calibration-sensitive; the matched direction is weak and the natural direction has a higher KL. We report both rather than choose the favourable one.

8 Ethics Statement

This work studies why models refuse and how that behaviour can be removed. It is dual-use: the same intervention that audits a safety mechanism can bypass it. We use public benchmark prompts, do not introduce a new attack, and do not include operational harmful content in this paper. Qualitative samples were reviewed locally and are not redistributed. We report the cost of the intervention (KL) rather than presenting a jailbreak as a free lunch. We encourage responsible disclosure and the use of these results to build defences that are not a single removable direction.

9 Reproducibility

The repository contains the runner, dataset construction, and table and figure generators under `src/`. The repository is available at `https://github.com/varadaraj772/neuron-subspace`. The matched, external, style and neuron experiments are reproduced by

```
python src/study.py --model qwen35_08b
python src/study.py --model lfm3_17b
python src/study.py --model r1distill_15b \
```

```
    --model-path models/r1
python src/study.py --model qwen35_4b \
    --model-path models/qwen35_4b \
    --dataset jbb --test-dataset same --force-layer 9
python src/make_tables.py
python src/make_figures.py
```

All splits are seeded and disjoint; the random controls use fixed seeds. The calibration-sensitivity runs (Table 3) are produced by omitting `-dataset jbb -test-dataset same -force-layer 9`, which estimates the direction on the natural split and tests it on all 100 JBB pairs.

Acknowledgments and Disclosure of Funding

This work was conducted independently, without external funding. The author declares no competing interests.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction, 2024.

Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwag, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramèr, Hamed Hassani, and Eric Wong. JailbreakBench: An open robustness benchmark for jailbreaking large language models, 2024.

Jianhui Chen, Xiaozhi Wang, Zijun Yao, Yushi Bai, Lei Hou, and Juanzi Li. Towards understanding safety alignment: A mechanistic perspective from safety neurons, 2024.

Thomas Marshall, Adam Scherlis, and Nora Belrose. Refusal in LLMs is an affine function, 2024.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal, 2024.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition, 2023.

Daniel Tan, David Chanin, Aengus Lynch, Dimitrios Kanoulas, Brooks Paige, Adrià Garriga-Alonso, and Robert Kirk. Analyzing the generalization and reliability of steering vectors, 2024.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering, 2023.

Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. The geometry of refusal in large language models: Concept cones and representational independence, 2025.

Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models, 2023.

Appendix A. Hyperparameters

Setting	Value	Notes
Calibration pairs (JBB)	50	direction estimate
Layer-selection prompts	96 / 48	AdvBench / Alpaca
Matched test pairs	30	held out
HarmBench behaviours	119	external
Style pairs	50	imperative vs question
Batch size	8	left padding
Generation	greedy	max 96 tokens
Dtype	float16	MPS
Seeds	0	splits, controls

Table 8: Hyperparameters for the main runs.

Appendix B. Redacted qualitative samples

We do not reproduce operational harmful text. The repository stores aggregate metrics, not model completions; qualitative samples were reviewed locally and are not redistributed. A typical pattern is: the baseline refuses ("I can't assist with that"); the ablated model complies, and the operational text is omitted here.

Appendix C. Full intervals

The main table reports 95% Wilson intervals. The calibration, external, style and addition tables report point estimates for space; the JSON artifacts contain the corresponding intervals. The intervals are wide for the 30-pair matched test (roughly ± 15 percentage points at 50%) and narrow for the 119-prompt external set.